Daming Yang

Curriculum Vitae

dyang@strw.leidenuniv.nl
dmyang42.github.io
dmyang42

PhD Candidate in Astronomy
Leiden Observatory, Leiden University

Education

2023–Present **Ph.D. in Astronomy**, *Leiden University*, Leiden, Netherlands
Expected completion: Summer 2027
Supervisor: Joseph Hennawi

2021–2023 **M.S. in Astronomy**, *Leiden University*, Leiden, Netherlands, *Cum laude*.
Supervisors: Joseph Hennawi (2nd year), Sjoert van Velzen (1st year)

2016–2020 **B.S. in Astronomy (the School of the Gifted Young)**, *University of Science and Technology of China*, Hefei, China
Supervisor: Jun-Xian Wang

2017–2020 **Minor in Computer Science**, *University of Science and Technology of China*, Hefei, China

Scientific Leadership

2026–Present **Co-lead, Euclid DR1 Key Project**
"Discovery of high redshift Euclid quasars", Euclid Consortium

Awards

2026 **Euclid STAR Prize – Student Award**, Euclid Consortium

Publications

As Primary Author

1. **D. Yang**, J. F. Hennawi, R. Bouwens, T. Kist, E. Bañados, et al., 2026, in press, *A dust-reddened quasar in a [CII]-luminous host at* $z = 6.55$ [Manuscript]
2. **D. Yang**, J. Hennawi, S. Bosman, F. Davies, R. Bouwens, et al., 2026, arXiv:2608.18212, in press, *Quasar Impostors: Two Extremely UV-Bright (*$M_{\rm UV} \approx -23.5$*) Reionisation-Epoch Galaxies Powered by Very Massive Stars*
3. **D. Yang**, J. Hennawi, F. Guarneri, J. Wolf, S. Belladitta, et. al., 2026, A&A, 711, A104, *Euclid: Discovery of 31 new quasars at* $6.6 < z < 7.8$
4. **D. Yang**, J.-T. Schindler, R. Nanni, J. Hennawi, et al. 2024, MNRAS, 528, 2679, *High-*z *Quasar Candidate Archive: A Spectroscopic Catalog of Quasars and Contaminants in Various Quasar Searches*
5. **D. Yang**, Z. Xie, J. Wang, 2021, RAA, 21, 99, *The Feasibility and Flexibility of Selecting Quasars by Variability Using Ensemble Machine Learning Algorithms*

As Contributing Author

1. Belladitta et al. 2026, A&A, 711, A150, *Euclid: A UV-faint quasar in a highly luminous star-forming host galaxy at* $z \approx 7.7$

2. Bouwens et al. 2026, A&A in press, arxiv: 2506.24128, *Beneath the Surface: $> 85\%$ of $z > 5.9$ QSOs in Massive Host Galaxies are UV-Faint*
3. Guarneri et al. 2026, A&A, 709, A241, *QUEST: Quasar unsupervised encoder and synthesis tool: A machine-learning framework for generating quasar spectra*
4. Davies et al. 2026, A&A in press, arxiv: 2603.10135, *Three Hundred Quasars from the Couch: A first look at high-redshift quasar discovery with SPHEREx*
5. Varadaraj et al. 2026, A&A, 707, A239, *Euclid: Discovery of bright $z \simeq 7$ Lyman-break galaxies in UltraVISTA and Euclid COSMOS*
6. Euclid Collaboration: Fu et al. 2026, A&A in press, arxiv: 2512.08803, *Euclid Quick Data Release (Q1): Euclid spectroscopy of quasars. 1. Identification and redshift determination of 3500 bright quasars*
7. Ezziati et al. 2025, A&A, 701, A282, *Owl-z: Bayesian tool for selecting $z \gtrsim 7$ quasars*
8. Fu et al. 2025, ApJS, 279, 54, *The CatSouth Quasar Candidate Catalog for the Southern Sky and a Unified All-sky Catalog Based on Gaia DR3*
9. Wang et al. 2025, MNRAS, 544, 687, *Luminous mid-IR selected obscured quasars at cosmic noon in SDSS Stripe82 II: spectroscopic diversity and broad Hα emissions*
10. Bañados et al. 2022, ApJS, 265, 29, *The Pan-STARRS1 $z > 5.6$ quasar survey II: 50 New Quasars at $5.6 < z < 6.5$*
11. Gloudemans et al. 2022, A&A, 668, A27, *Discovery of 24 radio-bright quasars at $4.9 \leq z \leq 6.6$ using low-frequency radio observations*

Observing Programs

Selected Observing Programs as Principal Investigator

2026 **ALMA, Cycle 13 2026.1.01562.S, Bands 6/7** 8.7 hours
Mining 1000 sq. deg for the UV-brightest reionization-era galaxies: an ALMA [CII] and dust census

2026 **JWST, Cycle 5 DDT 12761, NIRSpec/NIRCam** 8.1 hours
(co-PI with Francesco Guarneri, Julien Wolf)
Unlocking the Quasar Redshift Frontier: A JWST Benchmark for Early SMBH Growth and Reionization at $z > 8$

2026 **JWST, Cycle 5 SURVEY 11173, NIRSpec** Survey
(co-PI with Jan-Torge Schindler, Francesco Guarneri, Joseph Hennawi, Julien Wolf, Silvia Belladitta, Eduardo Bañados)
Pushing the Edge: A JWST Survey to Discover the Faintest Euclid QSOs at Cosmic Dawn

2025 **JWST, Cycle 4 GO 9180, NIRSpec** 94 hours
(co-PI with Joseph Hennawi, Timo Kist, Frederick Davies)
Ushering in the JWST Era of Precision Constraints on Reionization: A Survey of Faint Quasar IGM Damping Wings at $6.5 < z < 7.4$

2025 **ALMA, Cycle 12 2025.1.01312.S, Band 6** 10 hours
Revealing the Earliest SMBH-Galaxy Co-Evolution with newly discovered, low luminosity Euclid Quasars at $z \sim 7$

2025 **HST, Cycle 33 18112, WFC3** SNAP
(co-PI with Julien Wolf, Eduardo Bañados, Jan-Torge Schindler, Joseph Hennawi)
Beyond the Quasar Redshift Frontier: Uncovering Rapidly Accreting Supermassive Black Holes at $z > 8$ with HST/WFC3 and Euclid

Selected Observing Programs as Co-Investigator (major contributor)

2026B–2028A **Keck, KCWI/MOSFIRE** Large Program (16 nights)
PI: Joseph Hennawi; Keck Strategic Mission Support (KSMS)
Charting the Earliest Quasars and Brightest Galaxies during Reionization with Euclid and Keck

2026B–2029A **Gemini, GNIRS/Flamingos-2** Large Program (24 nights)
PI: Joseph Hennawi; GN-2026B-LP-214
Mapping the Luminous Universe at $z > 7$: A Gemini Spectroscopic Program to Confirm High-z Quasar and Galaxy Candidates from the Euclid Wide Survey

2024A–2026A **Keck, LRIS/KCWI/MOSFIRE** 26.5 nights
PI: Joseph Hennawi
A New Era of High Redshift Quasar Searches with Euclid

Allocations: 2024A, 2.5 nights (LRIS); 2024B, 6 nights (KCWI/MOSFIRE); 2025A, 6 nights (KCWI/MOSFIRE) + 2 nights (MOSFIRE); 2025B, 4 nights (KCWI/MOSFIRE); 2026A, 6 nights (KCWI/MOSFIRE).

2025B–2026A **Gemini, Flamingos-2** 60 hours
PI: Joseph Hennawi
A New Era of High Redshift Quasar Searches with Euclid

Allocations: 2025B, 30 hours; 2026A, 30 hours.

Selected Conferences and Talks

3–7 Aug 2026 **EREBUS Collaboration Meeting**
Ann Arbor, Michigan, United States — Contributed talk

27–31 Jul 2026 **Settling the Dust: Obscured AGN in Galaxy Evolution**
Lorentz Center, Leiden, Netherlands — **Invited focus session**

5–10 Jul 2026 **Euclid SWG-PU Workshop 2026**
Schloss Ringberg, Germany — Invited talk

25–29 May 2026 **Euclid Consortium Meeting 2026**
Barcelona, Spain — **Invited plenary talk**

12–17 Apr 2026 **Charting Cosmic Dawn in Copenhagen 2026**
Helsingør, Denmark — Contributed talk

23–27 Mar 2026 **Towards the Full Bloom of First Star, Galaxy, and Black Hole Formation Exploration — Tokyo JWST 2026**
Tokyo, Japan — Contributed talk

8–12 Sep 2025 **Massive Black Holes across Cosmic Time**
Cambridge, UK — Contributed talk

25–27 Mar 2025 **Euclid Consortium Meeting 2025**
Leiden, Netherlands — **Plenary talk**

18–21 Nov 2024 **Probing the Genesis of Supermassive Black Holes: Emerging Perspectives from JWST and Expectation toward New Wide-Field Survey Observations**
Kavli IPMU, Kashiwa, Japan — Contributed talk

26–29 Aug 2024 **Cosmic Dawn Revealed by JWST: The Physics of the First Stars, Galaxies, and Black Holes**
KITP, Santa Barbara, USA — Lightning talk

13–21 Jul 2024 **45th COSPAR Scientific Assembly**
Busan, South Korea — Contributed talk